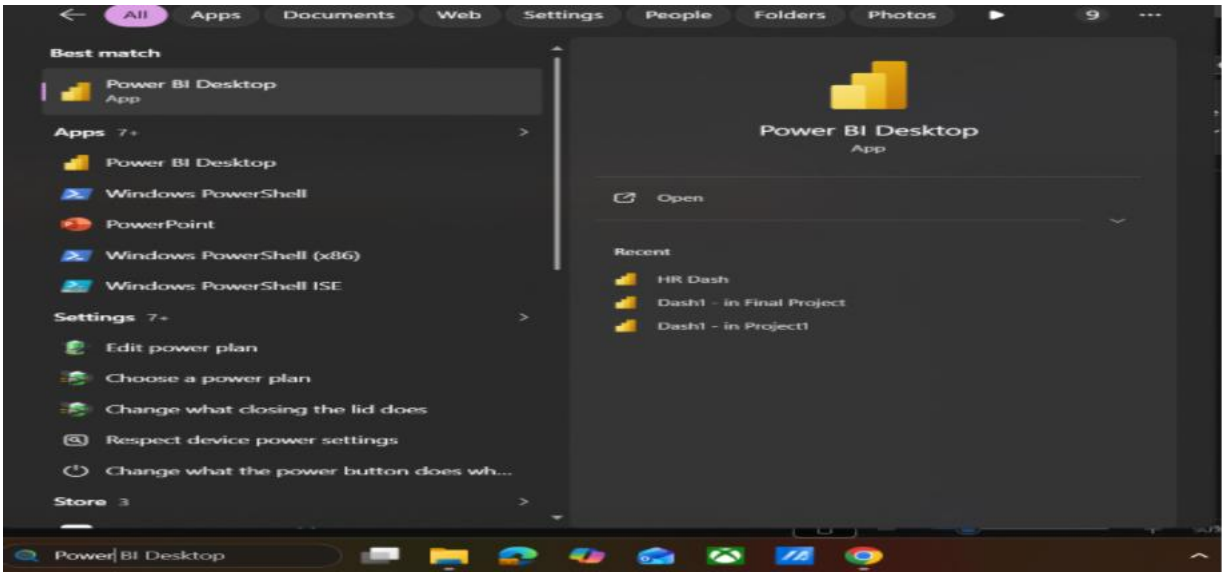
Practical 2: Report and Dashboard Creation

Objective: Create an interactive report using charts, slicers, filters, and publish it to Power BI Service.

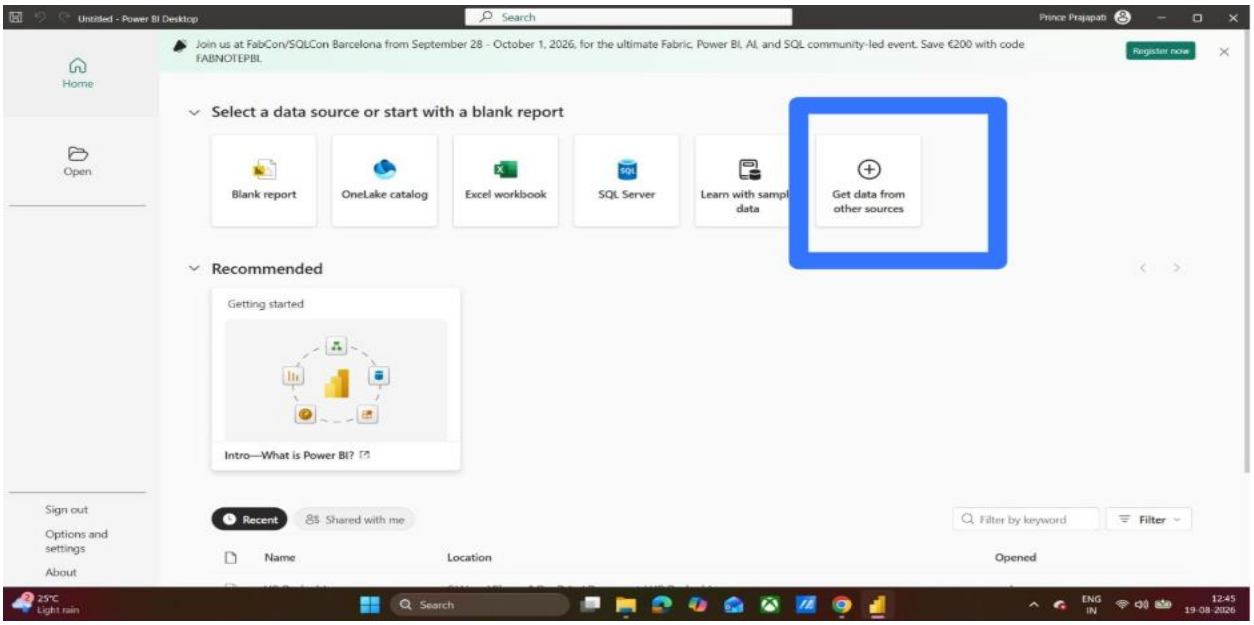
Step 1: Open Power BI Desktop

- Open the Search Menu, search for Power BI Desktop, and click the app icon to open the application.

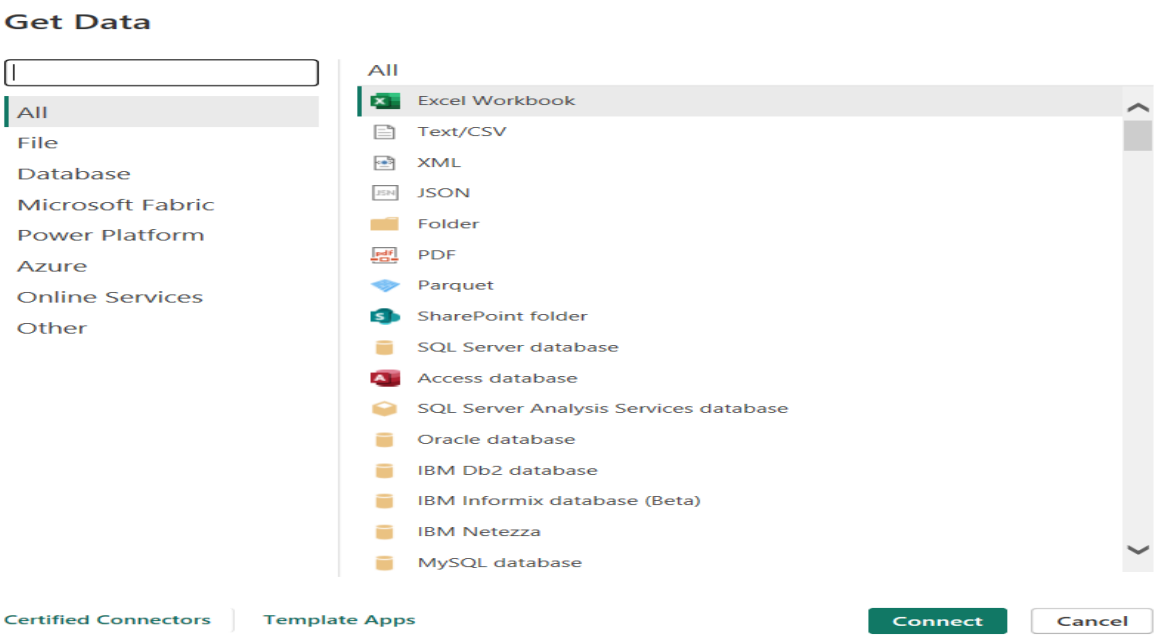


Step 2: Get the Data

- Go to the top ribbon tab in Power BI Desktop and click on **Home**.
- Click on the **Get Data** icon (or click the **Get data from other sources** tile on the welcome screen).

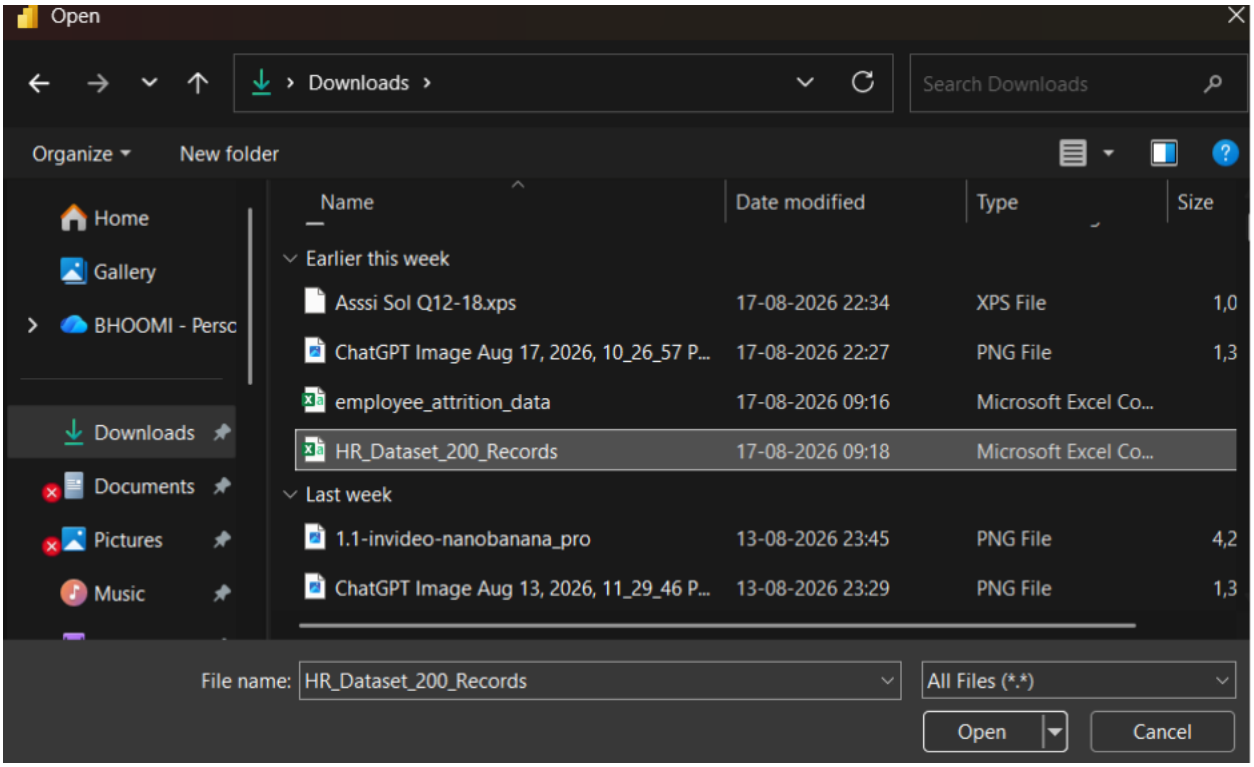


- Select your desired data connector format, such as Text/CSV or Excel **Workbook**, to proceed to the file selection dialog.



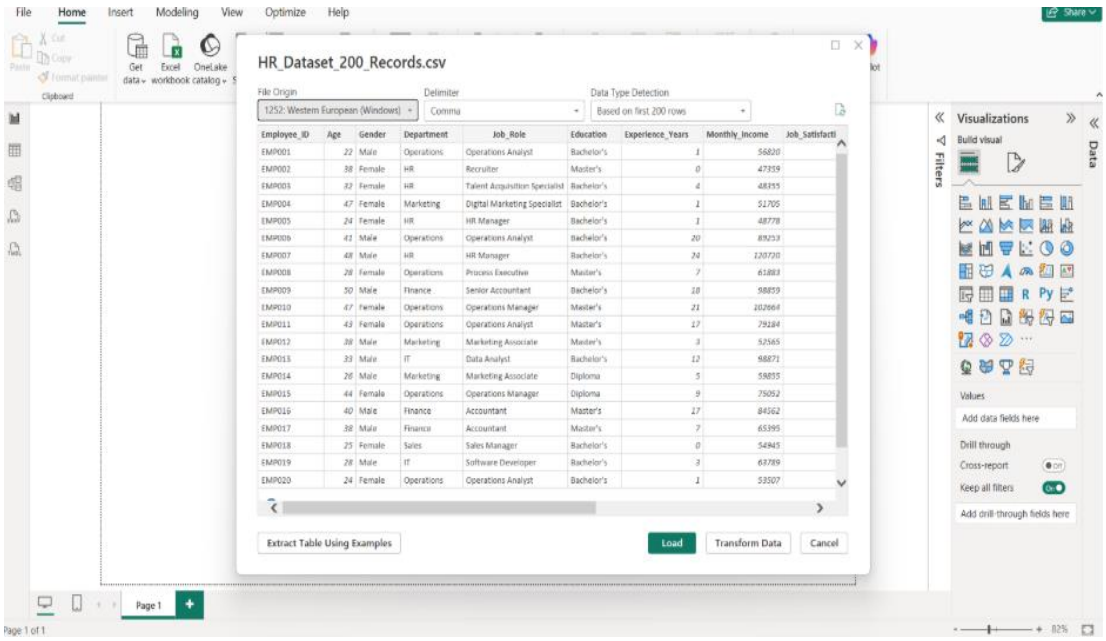
Step 3: Select Dataset

- Browse through your system directories in the Open file selection popup.
- Navigate to the specific folder path (e.g., `Downloads`) where your project dataset is stored.
- Click on the targeted file, such as `HR_Dataset_200_Records.csv`, to select it.
- Click the Open button at the bottom right corner of the window to proceed.

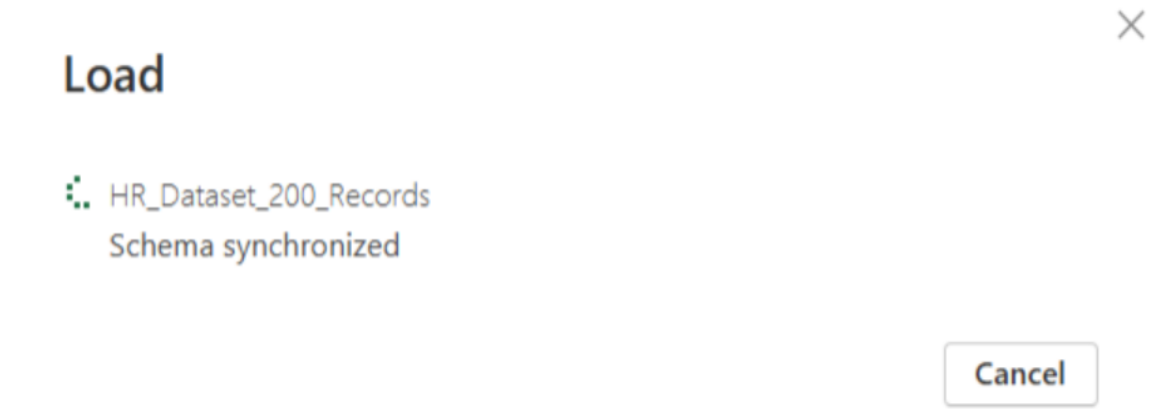


Step 4: Load Data

- Preview the dataset columns and rows displayed in the popup window.
- Verify that the table structure, column headers, and data types are aligned properly.



- Click the green Load button at the bottom of the window to import the dataset into your Power BI model.



Step 5: Create a Chart

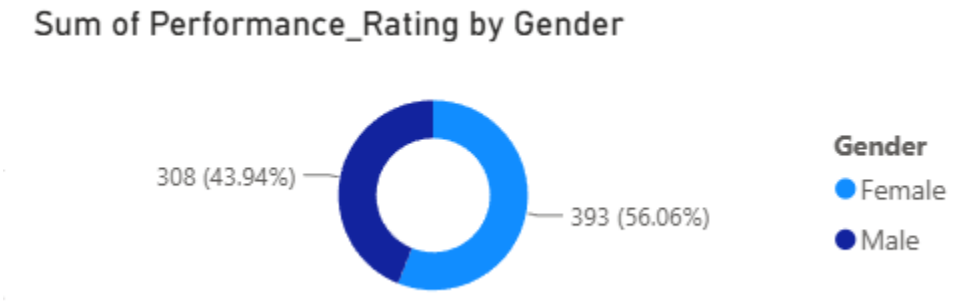
- Go to the Insert tab on the top ribbon and select Text Box. Type "HR Data Analysis" as your report title and place it at the top of the canvas.
- Select the Card visual from the Visualizations pane on the right side.
- Add three separate Card visuals across the top of the canvas to highlight the core aggregate metrics:
- Count of Department: Set field to `Department` (aggregated as Count) Displays 200.

- Sum of Performance Rating: Set field to `Performance_Rating` (aggregated as Sum) Displays 701.
- Count of Job Role: Set field to `Job_Role` (aggregated as Count) Displays 25.



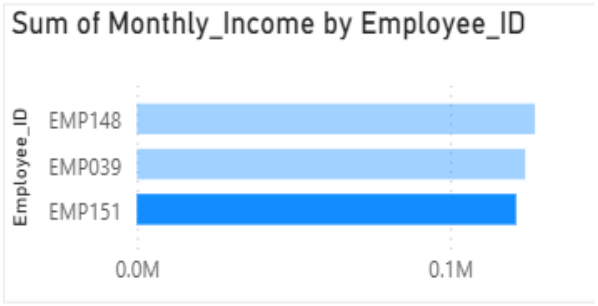
Step 6: Create Donut Charts for Categorical Breakdowns

- Select the Donut Chart visual from the Visualizations pane on the right side.
- Drag `Gender` into the Legend field and `Performance_Rating` (set to Sum) into the Values field. This creates a breakdown for Sum of Performance Rating by Gender (showing 56.06% Male and 43.94% Female).
- Select another Donut Chart visual from the Visualizations pane.
- Drag `Department` into the Legend field and `Performance_Rating` (set to Average) into the Values field. This displays the Average of Performance Rating by Department across Operations, Sales, IT, and HR.

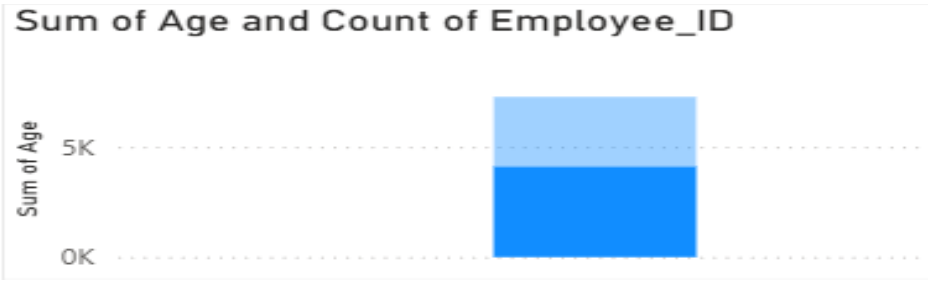


Step 7: Create Bar chart and Column Chart

- Add a horizontal Clustered Bar Chart to display Sum of Monthly_Income by Employee_ID.
- Add a Clustered Column Chart to show Sum of Years_at_Company by Job_Role.
- Insert a column visual representing Sum of Age and Count of Employee_ID.



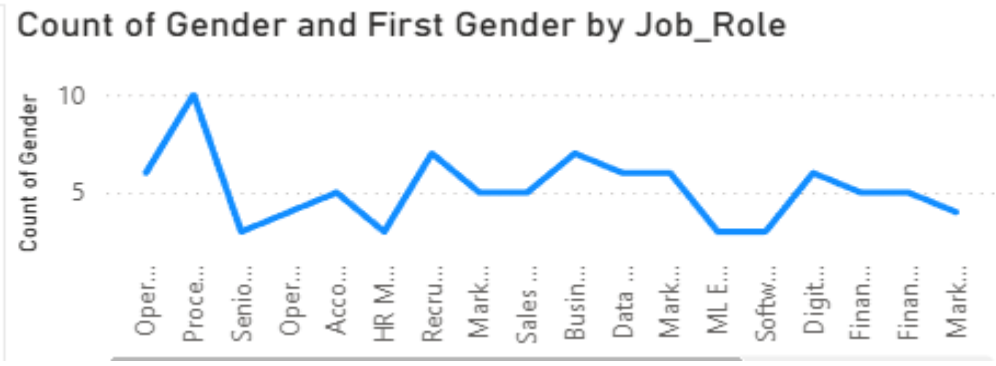
Bar chart



Column Chart

Step 8: Add Trend and Multi-Metric Visuals

- Insert a Line Chart displaying Count of Gender and First Gender by Job_Role.
- Add a combo/line chart showing Count of Education and Sum of Monthly_Income by Employee_ID.
- Insert a Treemap/Matrix visual to analyze Sum of Experience_Years and Sum of Monthly_Income by Employee_ID and Gender.
- Add a detailed breakdown donut visual for Sum of Experience_Years and Sum of Performance_Rating by Employee_ID.



Line chart



Treemap

Step 9: Configure Filters and Interactivity

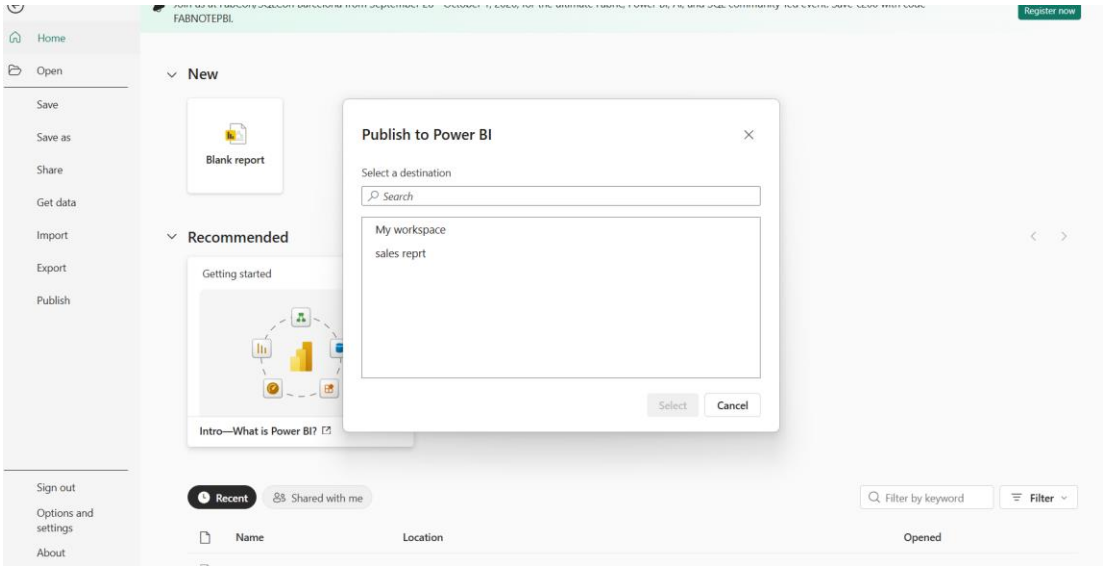
- Use the Filters pane to apply report-level and page-level criteria across all HR metrics.
- Enable cross-filtering so clicking any department, gender, or role automatically slices all related charts.

Step 10: Format Dashboard Layout

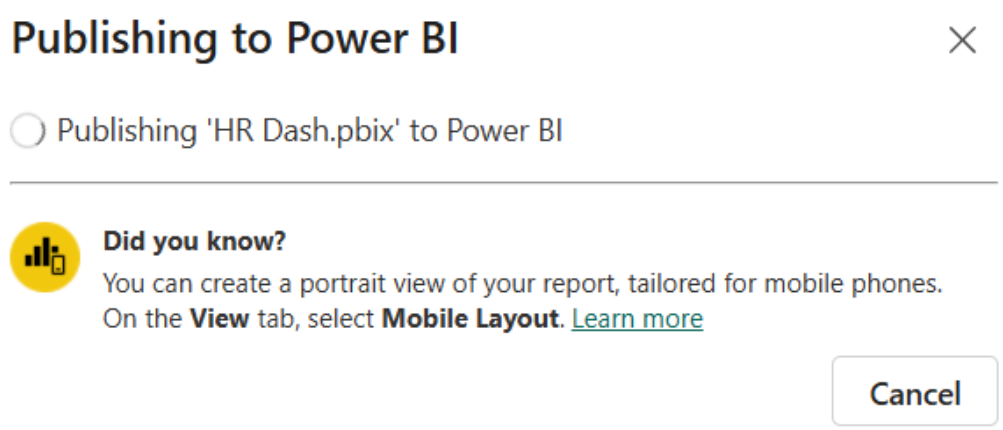
- Align visual containers symmetrically across a 3-tier grid layout.
- Customize data labels, legend positions, and chart borders for clear readability.

Step 11: Publish to Power BI Service

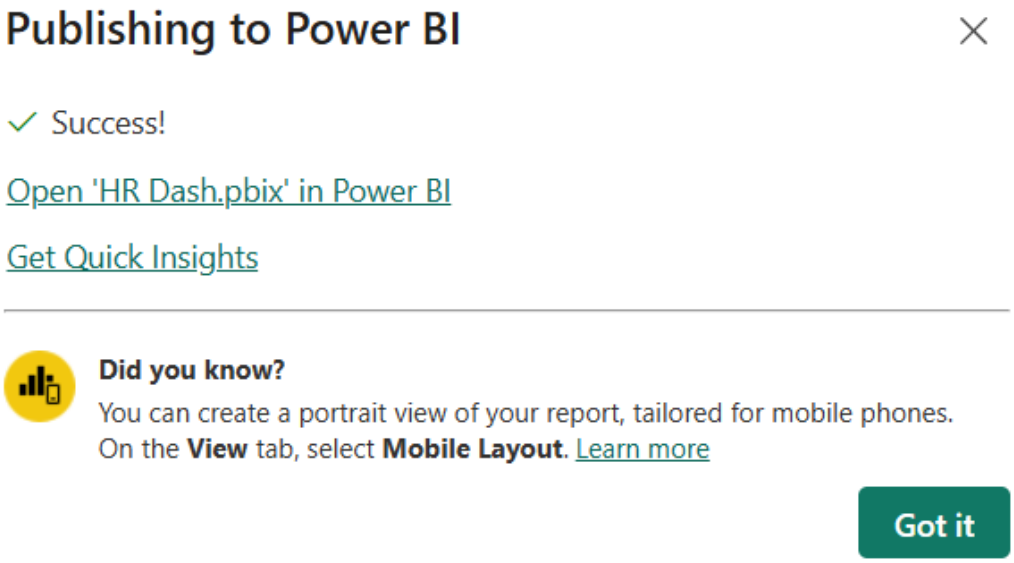
- Go to the **Home** tab on the top menu bar in Power BI Desktop and click **Publish**.
- Sign in using your Microsoft Power BI account credentials if prompted.
- Choose your destination workspace (e.g., **My workspace**) from the popup dialog and click **Select**.
- Wait for the file to upload and click **Got it** once the **Published (Success!)** confirmation window appears.



- Publishing **HR Dash.pbix** in progress.

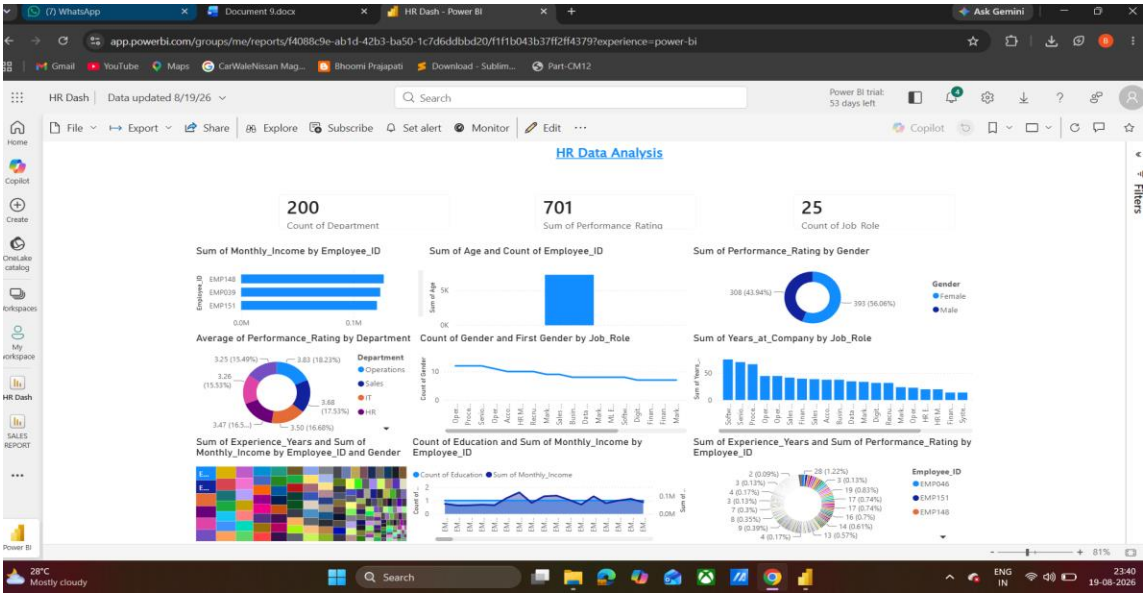


- Published (**Success!**)—click **Got it**.



Step 12: Verify in Power BI Service

- Log into Power BI Service (`app.powerbi.com`) in your browser.
- Open My workspace and click on the HR Dash report.
- Click charts to verify cross-filtering, KPIs, and visual interactivity.



- Report published and verified successfully in Power BI Service.

Result

The interactive **HR Data Analysis** dashboard was successfully created with KPI cards, donut charts, bar/column charts, and trend line visuals. All cross-visual filtering logic was validated, and the report was successfully published and verified in Power BI Service.